Task

You are a secret agent waiting at Edison Station, when someone runs past you, dropping a very old looking map at your feet.

Feeling very suspicious, you pick up the map and find that there is a strange decoder key beneath what appears to be an old copy of the railway stations in the area.

Some stations have been circled and appear to form part of a code, the stations appear in the 'Decoder Box' to the right.

You decide that this is something that has to be investigated and start your journey.

Remember to fill in the codes as you reach each station, you can change route at any of the stations that have a black circle, but you must attempt every question for every station that you pass.

Good luck!

Decoder	
Darwin Drive 6th letter of the answer	
Franklin Forest 3rd letter of the answer	
Mendeleev 1st letter of the answer	
Curie Corner Last letter of the answer	
Copernicus 4th letter of the answer	
Bunsen Road 1st letter of the answer	
Lister Lane 6th letter of the answer	
Washington Walk 3rd letter of the answer	
Hawking Street 5th letter of the answer	

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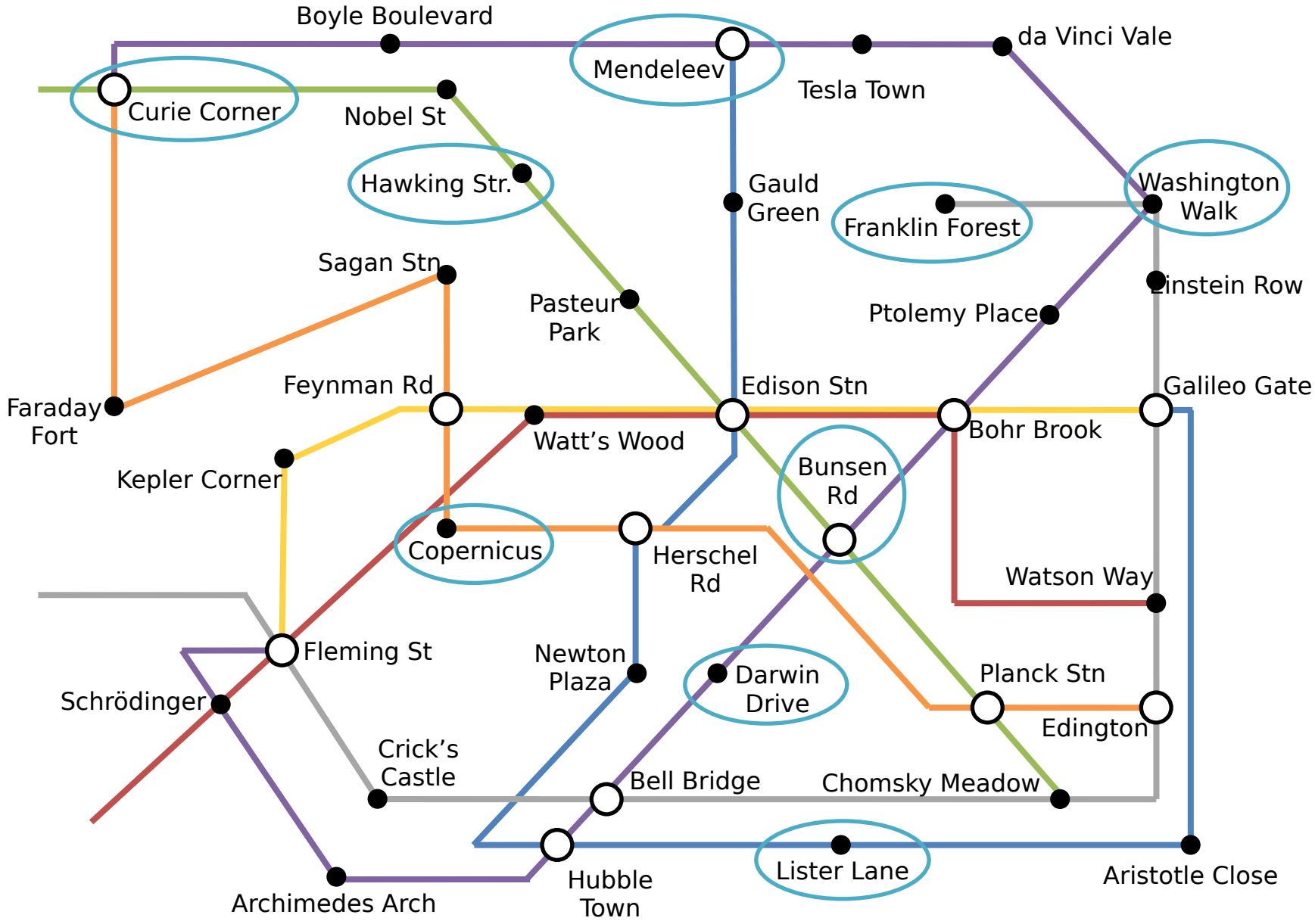

Boyle Boulevard
Mendeleev
da Vinci Vale
Tesla Town
Curie Corner
Nobel St
Hawking Str.
Gauld
Green
Franklin Forest
Washington
Walk
Sagan Stn
Pasteur
Park
Ptolemy Place
Einstein Row
Faraday
Fort
Feynman Rd
Edison Stn
Galileo Gate
Watt's Wood
Bohr Brook
Kepler Corner
Bunsen
Rd
Copernicus
Herschel
Rd
Watson Way
Fleming St
Newton
Plaza
Darwin
Drive
Planck Stn
Schrödinger
Edington
Crick's
Castle
Bell Bridge
Chomsky Meadow
Lister Lane
Aristotle Close
Archimedes Arch
Hubble
Town

Diagram sheet

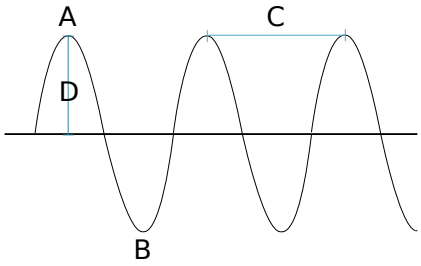


Figure 1

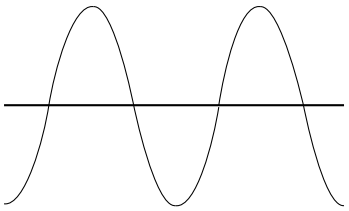


Figure 2

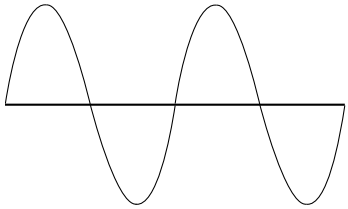


Figure 3

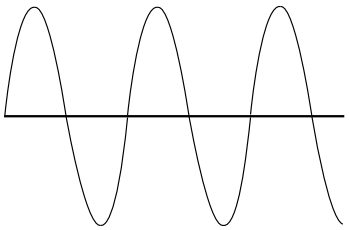


Figure 4

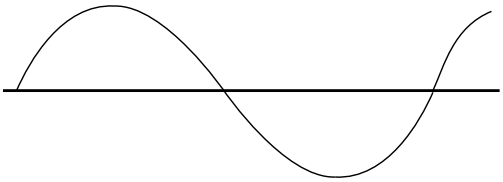


Figure 5

Diagram sheet

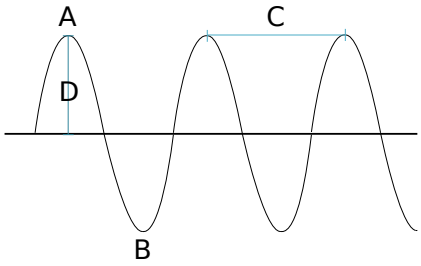


Figure 1

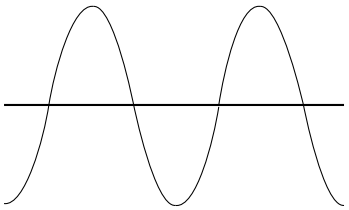


Figure 2

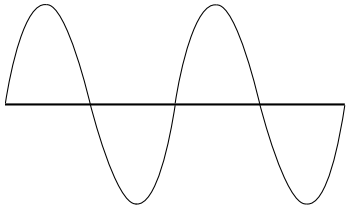


Figure 3

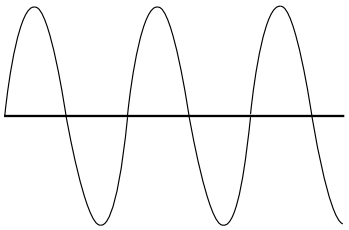


Figure 4

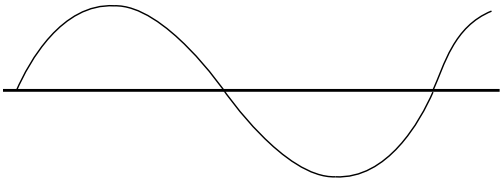


Figure 5

The quest begins

Answer all of your questions in the spaces provided.

Archimedes Arch	Are ultrasound wavelengths long or short?	
Aristotle Close	What is ‘B’ on figure 1?	
Bell Bridge	Look at figures 3, 4 and 5, which shows the highest pitched sound?	
Bohr Brook	What are the meanings of the two opposites ‘infra’ and ‘ultra’?	
Boyle Boulevard	What does ‘reflect’ mean?	
Bunsen Road	What is the technology called that sends sound waves underwater to detect shoals of fish, or survey the underwater landscape?	
Chomsky Meadow	What is meant by a vacuum?	
Copernicus	Sound waves transfer ……. to the particles that they touch.	
Crick’s Castle	What does 300 Hz mean?	
Curie Corner	What does the word oscillate mean?	
da Vinci Vale	What is the frequency of ultrasound?	
Darwin Drive	What type of sound wave is produced by dog whistles?	
Edington	Amplitude measures how much ……. is carried by a wave.	
Edison Station	What is the definition of frequency?	
Einstein Row	Which part of figure 1 represents wavelength, A, B, C or D?	
Faraday Fort	Sound waves have areas of high and low pressure. What type of wave is a sound wave?	
Feynman Road	Ultrasound waves are used for	
Fleming Street	When ultrasound passes from one substance to another what happens to some of the waves?	

Franklin Forest	Which of these produce a higher frequency sound a mouse or a cow?	
Galileo Gate	What is the part of the wave labelled ‘A’ on figure 1?	
Gauld Green	Changing the amplitude of a sound wave will also change the ..... of the sound.	
Hawking Street	Sound waves have areas of high and low pressure. What name is given to the area of high pressure?	
Herschel Road	What is the definition of wavelength?	
Hubble Town	What is ‘C’ on figure 1?	
Kepler Corner	What does a frequency of 50 Hz mean?	
Lister Lane	What is ‘D’ on figure 1?	
Mendeleev	Changing the frequency of a sound wave will also change the ..... of the sound.	
Newton Plaza	How many complete wavelengths are there in figure 1?	
Nobel Street	Sound travels fastest through solids, why?	
Pasteur Park	What type of wave is a sound wave, transverse or longitudinal?	
Planck Station	How do ultrasound beams break down kidney stones?	
Ptolemy Place	What word describes sound that is reflected?	
Sagan Station	What is meant by a ‘medium’ through which sound travels?	
Schröding er	What is the range of human hearing?	
Tesla Town	Name a medical use of ultrasound.	
Washington Walk	Which of these produce a lower frequency sound, a crying baby or a tractor?	
Watson Way	What unit is frequency measured in?	
Watt's Wood	Ultrasound has high ..... and	

	short	

The decoded word is **“superstar”**

Answer sheet

Archimedes Arch	Are ultrasound wavelengths long or short?	**Short**
Aristotle Close	What is ‘B’ on figure 1?	**Trough**
Bell Bridge	Look at figures 3, 4 and 5, which shows the highest pitched sound?	**Figure 4**
Bohr Brook	What are the meanings of the two opposites ‘infra’ and ‘ultra’?	**Infra – below, ultra - above**
Boyle Boulevard	What does ‘reflect’ mean?	**To bounce back**
Bunsen Road	What is the technology called that sends sound waves underwater to detect shoals of fish, or survey the underwater landscape?	**SONAR (Sound navigation and ranging)**
Chomsky Meadow	What is meant by a vacuum?	**Area with no particles**
Copernicus	Sound waves transfer ……. to the particles that they touch.	**Energy**
Crick’s Castle	What does 300 Hz mean?	**300 waves per second**
Curie Corner	What does the word oscillate mean?	**Vibrate**
da Vinci Vale	What is the frequency of ultrasound?	**Above 20 000 Hz**
Darwin Drive	What type of sound wave is produced by dog whistles?	**Ultrasound**
Edington	Amplitude measures how much ……. is carried by a wave.	**Energy**
Edison Station	What is the definition of frequency?	**The number of waves that pass a certain point each second.**
Einstein Row	Which part of figure 1 represents wavelength, A, B, C or D?	**C**
Faraday Fort	Sound waves have areas of high and low pressure. What type of wave is a sound wave?	**Pressure wave or longitudinal**
Feynman Road	Ultrasound waves are used for	**Cleaning, medical uses, sonar**
Fleming Street	When ultrasound passes from one substance to another what happens to some of the waves?	**Reflected back the way they came.**

Franklin Forest	Which of these produce a higher frequency sound a mouse or a cow?	**Mouse**
Galileo Gate	What is the part of the wave labelled ‘A’ on figure 1?	**Crest or peak**
Gauld Green	Changing the amplitude of a sound wave will also change the ..... of the sound.	**Volume**
Hawking Street	Sound waves have areas of high and low pressure. What name is given to the area of high pressure?	**Compression**
Herschel Road	What is the definition of wavelength?	**Distance in meters between two crests or troughs of a wave.**
Hubble Town	What is ‘C’ on figure 1?	**Wavelength**
Kepler Corner	What does a frequency of 50 Hz mean?	**50 waves that pass a point in a second**
Lister Lane	What is ‘D’ on figure 1?	**Amplitude**
Mendeleev	Changing the frequency of a sound wave will also change the ..... of the sound.	**Pitch**
Newton Plaza	How many complete wavelengths are there in figure 1?	**Two**
Nobel Street	Sound travels fastest through solids, why?	**Particles have a regular arrangement and are close together**
Pasteur Park	What type of wave is a sound wave, transverse or longitudinal?	**Longitudinal**
Planck Station	How do ultrasound beams break down kidney stones?	**They cause the kidney stone to vibrate which helps it break down.**
Ptolemy Place	What word describes sound that is reflected?	**Echo**
Sagan Station	What is meant by a ‘medium’ through which sound travels?	**Medium means substance.**
Schröding er	What is the range of human hearing?	**20 Hz – 20 000 Hz**
Tesla Town	Name a medical use of ultrasound.	**Scans, breaking down of kidney stones, cleaning teeth, cleaning medical equipment**
Washington Walk	Which of these produce a lower frequency sound, a crying baby or a tractor?	**Tractor**

Watson Way	What unit is frequency measured in?	**Hertz**
Watt's Wood	Ultrasound has high ..... and short	**Frequency, wavelength**